

Chapter 2

The Explanatory Role of Representations

. . . To unfold the secret laws and relations of those high faculties of thought by which all beyond merely perceptive knowledge of the world and of ourselves is attained or matured is an object which does not stand in need of commendation to a rational mind.

George Boole, *Laws of Thought*

Introduction

In this chapter I attempt to elaborate on the idea of levels of explanatory principles. Because one of the levels is the semantic level, I will be right in the middle of what is probably the second hardest puzzle in philosophy of mind: the puzzle of meaning. (The hardest puzzle, in my view, is consciousness, which probably is not even well enough defined to qualify as a puzzle.) I want to reassure the reader that I do not propose to solve the puzzle of what meaning is or how it gets into the head; my aims are somewhat more modest: to describe how the idea of the semantic content of representations is implicitly viewed within the field of cognitive science, and discuss why this view is justifiable. Although it is appropriate to raise this issue early, to provide a background for the more detailed analysis of cognitivist theorizing presented throughout the book, this strategy prevents me from using certain ideas and distinctions that will be developed later (notably, the notions of functional architecture and cognitive capacity). So bear with me; I will anticipate just enough of this material to lay out the story.

The first thing to do is introduce the idea that in order to express certain cognitive generalizations, we must refer to *representations*. This necessity is related to the point made in chapter 1, concerning certain generalizations being “under a cognitive description.” Here, though, I try to be more precise about what is entailed. In particular, I make the point that generalizations stated over the contents of representations are not mere *functional* generalizations in the usual sense. What is traditionally viewed as a functional description or a functional theory

(roughly, a theory that does not refer to physical properties of the particular system in question, only the way it operates; see, for example, Cummins, 1975; Fodor, 1968b; Haugeland, 1978; Putnam, 1973; Dennett, 1971), where it concerns cognition, needs to be refined. It is this refinement that forces us to raise, somewhat prematurely, the notion of computation, or symbol manipulation. In the refinement we will see that there are actually two distinct levels above the physical (or neurophysiological) level—a representational or semantical level and a symbol-processing level. Allen Newell (1982) refers to the first level as the “knowledge level” and to the second as the “symbol level.”

Having thus forewarned you of my intentions, I can now back up to where the discussion of explanation by appeal to generalizations left off.

The Appeal to Representations

Lawlike generalizations or explanatory principles that use different vocabularies can differ in several ways. Consider the following examples, each of which represents a fragment of a longer explanation. Account (1) says that a certain object accelerated at a meters per second per second because a steady force was applied to it that was equal to ma , where m is the mass of the object. Account (2) says that a certain neuron fired because a potential of v millivolts was applied along two of its dendrites and that it had been inactive during the previous t milliseconds. Account (3) says that the bit pattern in a certain computer register came to have a particular configuration because of the particular contents present in the instruction register and the program counter, and because the system is wired according to a certain register transfer protocol. Compare accounts (1) to (3) to account (4), which states that the computer printed the sequence of numbers 2, 4, 6, . . . because it started with the number 2 and added 2 repeatedly or because it applied the successor function repeatedly and doubled the value before printing it out; or with account (5), which says (as in our example in chapter 1) that the pedestrian dialed the phone number 911 because he believed it to be the emergency number and had recognized the urgent need for assistance.

In the first three accounts all terms refer to properties of objects within a closed system.¹ For example, the mass and force on an object,

1. In using the term *closed system* I do not mean to imply that the system has a well-defined physical boundary, or “skin”. The term is used in the systems-theory sense, where a system is defined in terms of a set of properties or parameters, together with some characterization of how they interact. The point here is that a closed system is one with no causal connections other than the set of variables that define the system. Although

the electrical potential across a dendritic membrane, the bit pattern in a computer register are all properties of a system that causally determine the sequence of states the system goes through. These accounts make no reference to entities or properties other than those that determine what is sometimes referred to as the *location* of the system in its state or phase space. Because these are deterministic systems (what Ashby [1956] calls “absolute” or “state determined” systems), closed to the outside except for effects explicitly recognized as inputs or outputs, such an explanatory account is also closed and complete.

Accounts (4) and (5) differ in an important respect: Both make substantive reference to entities or properties that are not an intrinsic part of their state description, for example, *numbers* or *need for assistance*. It is important to recognize that such terms do not refer to properties of the system whose behavior we are explaining (that is, the computer or the person doing the phoning) nor necessarily to properties causally connected to such states or to the system’s behavior, that is, the behavior of the system cannot be explained in terms of a chain of causal laws connecting such extrinsic entities as those mentioned in accounts (4) and (5) to the current state of the system. The reason for this has already been discussed in chapter 1, that is, the causal chain would have to contain an arbitrarily large disjunction of possible physical causes, since the relevant connections between the extrinsic events and the current state includes such relations as: being told about them by someone, reading about them, inferring from what one has seen or heard, and so on. Neither numbers nor the anticipation of help can in any sense be viewed as *causal inputs* to the system. Numbers are abstract mathematical objects; and the successor function is defined only over these abstract objects. Similarly, the content of anticipations or goals are possible future states of affairs; none can enter directly as causes in the usual sense. The case is even clearer when we find people’s behavior being a function of such nonexistent things as Santa Claus, the pot of gold at the end of the rainbow, a hoped-for raise, or beliefs that are false (for instance, in the example in chapter 1, the perceived *need for assistance* could be an illusion: the pedestrian could have happened upon a rehearsal of a movie scene).

As materialists we must ask how behavioral regularities such as those captured by statements that mention extrinsic but causally unconnected entities (or perhaps nonexistent objects) are possible in a world governed

no segment of the universe is truly closed, it is often possible to define a set of properties and a configuration such that the behavior of the configuration is determined almost entirely from its initial state. We can then extend the notion to any configuration that has well-defined interactions with extrinsic properties by viewing the interactions as occurring through inputs and outputs to the system, which then become part of the system description.

by physical laws. How is it possible for properties of the world to determine behavior when the properties are not causally related in the required sense to the functional states of the system, which is what we seem to be claiming when we say, for example, that what determines the person's behavior in rushing to the phone is something like his anticipation of, or desire to obtain help? This conundrum is sometimes called "Brentano's problem," after the psychologist who first appreciated its force. Brentano saw no way out of the puzzle except by accepting that mental events are not governed by physical laws but by their own mental principles. The answer I ultimately give to this puzzle is that the causes of the behavior are not literally numbers, anticipated future events, or other "intentional objects" but rather some physically instantiated *internal representation* of such things, that is, a physical code or a symbol.

To introduce this discussion it should again be emphasized that the notion of representation is necessary only in the context of explanation; it is needed to state generalizations concerning the behavior of systems under certain descriptions. (Recall that explanation is relative to particular descriptions of the behavior of a system.) Thus, if under a particular description of a system's behavior, a physical, neural, or purely functional account captures all the relevant generalizations hence serves the explanatory function, then appealing to representations is not essential. Included in examples of such heuristic uses of representation-talk are, for example, explaining the operation of a television set, watch, or thermostat by referring to certain properties of the device as representing, say, time or color or temperature. In such cases the representational character of certain properties enters in only in order to connect the operation of the device to the intentions of its designer. What is crucial here is that we do not need the notion of representation to explain how a device works, only to explain how it performs the function intended by its designer. In other words, the content of the representations—or, what the properties in question actually represent—is not part of the explanation itself. We need not say that the system does such and such *because* it has a certain representation, though frequently we use such language to express the reason why the system was designed in a particular way. Here, the test is that, by viewing the device as transforming physical patterns of inputs into physical patterns of outputs, we lose no explanatory power, inasmuch as everything that must be accounted for—all relevant generalizations—can be captured in an account that makes no mention of what various properties and states represent. This is in marked contrast with the case of explaining genuine representational or intentional processes.

To see what makes the crucial difference between mere design de-

scriptions (what Dennett [1978] refers to as descriptions from the “design stance”) and genuine cases of epistemic states, consider the following example, in which the important generalization cannot be captured unless we refer to the semantic content of the representation.² Suppose I wish to explain what I am doing at this moment as I sit here in front of my computer terminal, alternately typing these sentences and thinking about other things. I look out over the Stanford campus below, and my mind wanders. I think about the Santa Cruz mountains behind me and wonder why I didn’t go for a walk after lunch instead of confining myself to my study. Clearly, my current behavior (including writing this paragraph) is caused by my current thoughts and goals. They include the goal of having a completed chapter and the thought that there are walking trails in the hills behind me. If there is any sense in which this behavior is *caused* by the nonexistent, completed chapter or by the hills, it is an entirely obscure sense of causation. Plainly, what is going on is, my behavior is being caused by certain states of my brain. Yet—and this is the crux of the problem—the only way to explain why those states caused me to type the specific sentences *about* walking, writing, the mountains, and so on is to say that these states themselves are in some way related to the things referred to (writing, walking, mountains).

This is not simply a case of the explanations having to be more complicated or less comprehensible if the content of the representation is not mentioned; the relevant generalization or connection to my behavior, viewed under the appropriate description, cannot even be expressed. My brain states are not, as we have noted, causally connected in appropriate ways to walking and to mountains. The relationship must be one of content: a semantic, not a causal, relation. Brain states cause certain movements. If these movements are viewed as members of the equivalence class of behaviors described as “writing a sentence about walking in the Santa Cruz mountains” the brain states must be treated as embodying representations or codes *for* such things as walking and hills. In this way the behavior can be seen as rationally connected to the representational content of these codes by certain rules (logical, heuristic, associative, etc.). That must be the case in part because the meaning of my words derives from the semantics of my mental states, from my intention to mean something or other by expressing them. If the mental states that caused *those* sentences to be produced were about

2. Here, the notion of content is roughly that of what the states are about or what they represent. It is in itself not entirely unproblematic, though the general idea is clear enough. In End Note 2, I discuss some of the distinctions that have to be kept in mind when talking about content.

nothing (or were about something quite different), then my producing the sentences would be just like a tape recorder's or photocopier's producing sentences, it would not constitute the making of an utterance; therefore, an explanation cast in these terms would not explain the event under the description that something was being asserted. Presumably, it is under *that* description that we will find my keyboard pressings to be explainable in terms of certain of my experiences (that is, related in terms of some system of regularities or generalizations).

On the other hand, in the case of such artifacts as watches, the semantic interpretation is gratuitous, since the set of movements corresponding to the "interpreted" behavior is coextensive with, or type-equivalent to, the set of movements corresponding to the physical description of the behavior. Whether you call something the position of a lever or the time of day is determined merely by whether you are interested in conveying to someone the intended use of the device; it has no bearing on capturing generalizations about the possible behaviors of the device, since the device *exhibits no other behavior* that falls under generalizations involving time of day as an open-ended category (a category realizable in arbitrarily many distinct physical forms). The same is true of a description of more complex devices, for example, a television set. In terms of the discussion above of explanations being relativized to phenomena "under descriptions," the two forms of explanation (say, of the behavior of a clock or a television set, or any other artifacts designed for a specific function), the physical account and the representational account end up explaining their behavior under a coextensive taxonomy; hence, one captures no generalizations missed by the other. On the other hand, the two ways of explaining certain human behavior capture extremely different generalizations.³

Representational and Functional Levels

The burden of these examples (as well as the example in chapter 1) is to suggest that it is not enough to give a *functional* description of mental processes; we must also discuss the content of certain representational states. If, however, the content makes a difference to behavior, is it not, by definition, also a functional difference? There is a sense in which functional and representational states might be viewed as identical except for the way we choose to describe them (and some, notably Stich [1984] take precisely this view). If that were the case, one might

3. In End Note 1, I discuss some indicators of when the attribution of representational content or semantics to functional states of a system plays an ineliminable part in the explanation.

think nothing could be gained by adding the semantic vocabulary to the account; that is, nothing is gained by mentioning what states represent. While it is true that differences in content always result in functional differences, the converse may very well not be true. Thus the question we need to ask is whether we can have differences in functional states *without* there being differences in content, or without there being differences in what is being represented. To answer this question we need to be somewhat more specific about how representational states and functional states are related. What I shall try to do at this point is anticipate some of the discussion to come in the next several chapters, in order to deal with the content-attribution issue here.

The view I take in chapters 3 and 4—with a few exceptions to be mentioned when the topic comes up for discussion, and which is fairly widespread in cognitive science—is that, to be in a certain representational state is to have a certain symbolic expression in some part of memory.⁴ That expression *encodes* the semantic interpretation and the combinatorial structure of the expression encodes the relation among the contents of the subexpressions, much as in the combinatorial system of predicate calculus. The details are very much in dispute and, I might add, subject to empirical investigation; but the general picture seems to be widely accepted. This picture is discussed in the following chapters, where some of its aspects are defended.

The reason there must be symbolic codes, of course, is that, because they are instantiated as physical patterns, they can enter into causal relations. Unlike semantic contents, tokens of symbols can actually cause the system to behave in certain ways. Now, if there is a unique symbolic expression corresponding to each content, one might expect

4. Within philosophy, this corresponds to the view championed by Fodor (1980c), that to have a certain propositional attitude that is, to “believe” or “want” or “fear” *P* is to be in a certain computational relation to a representation of *P*. Others, such as Davidson (1970) and Geach (1957, 1980), insist that it is unnecessary to assume that something (an internal property or a symbol) corresponds to *P* to explain how people can be in the state of “believing *P*.” All we need, according to this view, are certain states of an organism which function in a particular way—namely, in a way correctly described (from the outside) as “*P*-believing.” This is a kind of adverbial theory of intentional states which simply pairs functional states with belief (or other propositional attitude) descriptions without the step of positing any articulated substates or symbols that are the representations of *P*. This idea is discussed briefly in chapter 3, where I argue that within the context of explanation it is necessary that a functional theory take a certain articulated form, that it have a symbolic level or a level of syntactic structure. The regularities of a system with arbitrarily many functional states cannot be captured in a finite manner without some language-like combinatorial mechanism that ties together the systematic features of the set of state transitions. Indeed, this is a primary reason for a computational model appearing to be the appropriate one for cognitive processes.

functional states and representational states to once again be in a one-to-one relation; hence, the ascription of content still does not gain us any explanatory leverage—it remains merely heuristic.⁵ That need not be the case, for two reasons:

(a) There may be synonymous expressions—sets of codes with the same semantic content. Such codes might be functionally, but not semantically, distinguishable. For example, one might be structurally more complex than another and, consequently, might take more time to process or lead to certain systematic errors—predictable according to certain functional principles at what is referred to below as the “symbol level.” (The problem of distinguishing differences in content from differences in structure remains to be solved. According to Putnam (1975,1981), it may be one of those puzzles better left alone.)

(b) Merely possessing a certain symbolic expression that encodes semantic content is, by itself, insufficient to produce behavior. Another part of the story remains to be told; it is a part that has been of much greater concern to computer scientists than to philosophers or psychologists (see, however, Newell, 1972,1973b). What is needed is a set of mechanisms to make the system run or, as is sometimes said, “interpret” the symbols. It is because of these mechanisms that the symbolic expressions do not exhaust what we have informally been referring to as the “functional state” of the system.

The set of mechanisms needed is approximately what I refer to below as the “functional architecture” of the system (though I place severe constraints on what qualifies as functional architecture). It includes the basic operations provided by the biological substrate, say, for storing and retrieving symbols, comparing them, treating them differently as a function of how they are stored, (hence, as a function of whether they represent beliefs or goals), and so on, as well as such basic resources and constraints of the system, as a limited memory. It also includes what computer scientists refer to as the “control structure,” which selects which rules to apply at various times. (See the brief discussion of the problem of control structure at the end of chapter 3.)

Some of these mechanisms may be quite rigid, whereas others may be alterable in various ways by the environment. For example, retrieval

5. Even in this case, the word *merely* overstates the case. Content ascription may still be crucial insofar as there may be no way to discover the functional states—no independent motivation for positing equivalence classes of physical states—than by treating the system as an “intentional system” and in this way inferring its beliefs, goals, and other semantic contents. After all, there are infinitely many ways to map physical states onto functional ones. Presumably, the correct way is the one that captures generalizations stated over the appropriate vocabulary, which, in the case of cognition, is extremely likely to be the intentional one.

and control processes may be sensitive to anything from intoxication and sleep to the association strength among codes, which reflects, say, the repetition or the temporal proximity of token occurrences of their referents. Of course, the mechanisms themselves are not sensitive to the *content* of incoming information, since, by hypothesis, semantic content is precisely what is encoded in terms of the symbolic codes (this point is quite central in later discussions, where I talk about the "cognitive impenetrability" of the basic mechanisms).

The importance of the distinction between semantically interpreted symbolic codes and functional architecture arises when we relate the functional description to the representational one. When we claim that functional states are semantically interpreted (that is, they have representational content), we are referring only to the symbolic code part of the states, not the functional architecture. Differences in functional architecture, though they affect behavior, are not considered related to differences in content; the representation can remain the same despite changes in functional architecture. What that means is: If we individuate states according to their content, then two systems in exactly the same representational state (all the same goals, beliefs, fears, desires, and so on) can nonetheless still be functionally distinguished. That's because behavior is determined by more than the mere content of all one's beliefs, goals, and so forth. Behavior is also determined by the functional architecture, including the control structure and all available mechanisms.

Take a simple case. Two people whose beliefs and goals are all the same may nevertheless be functionally distinguishable if they have, say, slightly different memory-retrieval mechanisms, short-term memory capacities, degrees of alertness, or, possibly, different reinforcement histories, insofar as these cause differences in some aspect of the functional architecture rather than the content of certain beliefs (assuming reinforcement indeed has such an effect, as is claimed by behaviorists though denied by others, for example, Brewer, 1974). Two people who differ in this way might take different amounts of time on different classes of problems; they might make predictably different errors, find various tasks easy or hard, and so on. That is why psychologists devote so much time to studying failures; failures provide evidence of humans' functional architecture. On the other hand, people in artificial intelligence spend considerably more time studying expert performance and determining what knowledge these experts bring to bear on problems with which they are fluent—because in artificial intelligence one is more or less stuck with a relatively limited class of functional architectures, namely, those of available computers equipped with available high-level languages.

It thus appears that when we provide the kind of functional explanation provided in cognitive science—in terms of both semantically interpreted symbol structures and properties of the functional architecture—we find that many functionally distinct systems have the same representations. In other words, precisely the same representational state may lead to behaviors that differ in predictable ways. Thus generalizations can be captured by referring to the content of representational states that differ from those captured by referring to functional mechanisms. Generalizations expressible in terms of the semantic content of representations are referred to in this book as “semantic-level generalizations,” whereas generalizations expressible in terms of functional properties of the functional architecture are referred to as “symbol-level generalizations.” The latter term is taken from Newell (1980,1982), who uses the term “knowledge level” where I use “semantic level.” Although “knowledge level” conveys the correct idea to people in artificial intelligence, it has the unfortunate property of failing to distinguish between knowledge and belief; further, it gives the appearance (perhaps incorrectly) of leaving out such other representations as goals. That is why I will continue to use the term “semantic-level generalization.”

Representational Content as Defining a Level of Description

Although the need to use a special class of terms (especially terms that refer to semantic content) may surprise some, the situation here is no different from one we have already encountered. The principle that leads us to postulate representational states (individuated by their content) that are distinct from functional states is exactly the same as the principle that leads us to postulate functional states that are distinct from physical states. In both cases we want to capture certain generalizations. We discover that in order to do this, we must adopt a new vocabulary or taxonomy; hence we find ourselves positing a new, autonomous level of description.

Recall that our reason for abandoning a biological vocabulary in describing cognitive processes is that we found there are many generalizations (at least in such central processes as thinking and problem-solving) we have reason to believe correspond to arbitrarily large disjunctions of sequences of brain states, that is, there is multiple realization of functional states as neurophysiological states.⁶ We therefore concluded

6. Even if some functional states or properties did correspond to particular neurological properties on *every* occasion, it would be a *discovery* of some importance. It would be a “discovery” precisely because there would have been two independent sets of considerations or methodologies that happened to converge on one of their categories, a con-

that biologically distinguishable brain states are not identical to functionally distinguishable states, consequently, functional generalizations cannot, in general, be captured in a finite neurophysiological description.

It is the fact that certain arbitrary sets of physical properties have something in common at the functional level, which cannot be captured in terms of a finite description at the physical level, that leads us to postulate the existence of a new level with a distinct taxonomy or vocabulary. In this case it is a purely functional vocabulary in which theoretical constructs are identified by their role in explaining or generating the behavioral regularities in question (for example, the symbolic terms that appear in any computational or information-processing model of a psychological process). When we have principles of operation that cannot be stated within a certain vocabulary—but that *can* be captured in another, more abstract (here, *functional*) vocabulary to which the terms of the first vocabulary stand in a multiple-realization relation—we have a *prima facie* case for the existence of an independent level of description. This is exactly the situation we find ourselves in with regard to regularities expressible in semantic terms. That was the idea behind the analysis above of such examples as the one in which a pedestrian who witnesses an accident dials the emergency telephone number, or in which we consider how my writing the particular sentences I wrote might be explained. On the basis of these considerations we can conclude that the representational, or semantic, level represents a distinct, autonomous level of description of certain kinds of systems. The notion that there is something special about certain systems, that requires their description in terms of representations or representation-governed processes is only beginning to be appreciated by those concerned with the foundations of cognitive science (Fodor, 1980c; Dennett, 1978, 1982; Newell, 1982), though it is implicit in nearly all the work done by cognitive psychologists and researchers in artificial intelligence.

One way such an implicit view manifests itself is in the general acceptance of the assumption that there are certain *principles* constraining transitions among states individuated in terms of content and that these principles do not apply to states individuated by function

vergence that is a contingent empirical fact, not a logical necessity (indeed, it might not even be true in another species or for other categories). Further, even if, as seems unlikely at present, there were an abstract physical description that in some sense always preserves the functional types, this would still not provide a basis for arguing against the existence of an autonomous, functional level, since one would not be able to read off the functional generalizations from the physical description alone, inasmuch as the *same* physical system has many functional descriptions, depending on what we call its inputs and what its outputs, that is, depending on the description under which we view its behavior. Here, the abstract, physical description would still have to be after the fact and would have to be relativized to at least some prior functional considerations.

alone. Despite the acceptance of this assumption by virtually everyone who uses what philosophers call intentional terms (who talks about beliefs, goals, fears, and other “propositional attitudes”), such principles are understood only roughly. Chief among them is the principle of *rationality*. To the extent we are justified in appealing to such a principle that is, to the extent that, by using this principle, we can give a general account of certain classes of regularities and thereby render coherent a range of behaviors that would otherwise just be an unprincipled collection of empirical generalizations—to that extent we are justified in claiming to have discovered a distinct level of analysis beyond the functional level. What I am claiming is that the principle of rationality (and perhaps other principles that are stated over the content of representations) is a major reason for our belief that a purely functional account will fail to capture certain generalizations, hence, that a distinct new level is required. It is because of this belief that the principle of rationality figures so prominently in the writings of people intent on understanding intentionality. It is also due to this belief that practitioners of cognitive science who speak of goals and beliefs take rationality for granted. For example, we take it for granted that when we say someone does *A* because they have goal *G*, and because they believe that doing *A* will help them achieve *G*, we are, in fact, giving a *principled* explanation (rather than merely an *ad hoc* description) of what happened.

For example, in explaining how a subject was able to assign the correct anaphoric reference to a certain pronoun (say, the italicized pronoun in the sentence “The doctor did not try to help the nurse because *he* needed the practice”), we do not hesitate to refer to what the subject believes (or “knows”) about the world, including beliefs about stereotypical sex roles, which, for example, would explain the difficulty many people have in interpreting the sentence. Not only do we unhesitatingly give such explanations, we tacitly assume that they are principled. That is, we do not feel compelled to add something like, “and if the person *was acting rationally at the time*”; it is assumed to be part of the “normal background conditions.”

It is not simply that rational action occurs frequently; rather, it is that there is something special and nonarbitrary about concluding *Q* from *P* and “*P* entails *Q*” that makes it the unmarked case. If people have the beliefs we attribute to them—but consistently arrive at consequent beliefs whose content bears no such privileged relation to these beliefs (say, if a man asserts that he is Napoleon whenever he learns it is raining)—then, that *would* require a further analysis and explication, precisely because it lies outside such a principled relation. I will return to this issue by a somewhat different route.

Levels and Constraints on Realizability

Throughout our discussion of levels up to this point I have referred to the need to capture generalizations. Indeed, this is the main argument for the existence of levels, that is, that there be valid generalizations at one level that are not expressible at a lower level. There is another, closely related way of looking at this notion, however, that can help illustrate the role played by the principle of rationality while simultaneously relating the idea of a level to constraints rather than actual performance. This approach is related to the notion of *capacity*, one of the main points of focus in chapter 7. What I suggest is that the existence of a new level can often be recognized by the presence of constraints on the behavior of a system over and above the constraints that can be expressed in terms of principles currently available. I make the point that the purely functional constraints imposed by the functional architecture are compatible with the existence of systems whose behavior clearly falls outside what is natural for a “cognizing” system like us, hence that the explanation of such constraints, if they are principled ones, must be provided at another level. There is an interesting analogy here with biology, which I will take a page or so to examine first, to show how the existence of genuine explanatory levels provides for a kind of emergence of constraining principles.

One way in which this emergence is exhibited is that at one level of description, the principles of operation of a system may be compatible with a much wider range of behaviors than actually occurs. For example, the laws of physics are compatible with the existence of all the observed types of creatures as well as an indefinite number that do not exist. Why does only this subset exist? One answer might be that it’s just a fluke; the initial state of the universe had to take some form, and the form it took led to the evolution of precisely this subset of creatures, and no others. Another answer might be that members of the existing subset had a survival advantage; but that is no explanation unless one can identify particular properties of the existing creatures and can explain how those properties increase the chances of survival of the group, by what mechanisms the property is realized and propagated, especially if it is a functional property, such as a high rate of reproduction. A more interesting answer to the question of why only certain patterns exist would be possible if there were a set of principles that characterized the existing patterns as a class. For example, if biology picks out a natural domain of inquiry—that is, if *living thing* is a natural kind—then there will be principles that apply to all entities in that domain and only entities in that domain. In other words, if some phenomenon falls within that domain, it must obey the laws of biology. On the assumption that this or something like it is true of biology, there may

be principles living organisms cannot fail to follow and still qualify as living organisms. If some configuration of atoms failed to obey such principles, the configuration would automatically fall outside the domain, hence would be subject to *no* biological principles. For example, universal biological laws may exist that constrain the possibilities for tissue structures, say, laws of genetics or embryology. Systems that do not obey such laws simply would not be living organisms, not by definition but by an empirical theory, because biology claims that living organisms as a class must follow such principles. Consequently, whereas physics provides a set of constraints on what is physically possible, there may be cases for which one can give a principled *explanation* of why a certain class of these possibilities do not occur, by appealing to principles at another level, say, to principles of biology, genetics, embryology, or evolution. Thus, in cases where there are such principles (and there may not always be such principles; as I state above, the exclusion might merely be an accident of the initial state), the existence of another level (and, typically, a different nonreducible science) is usually indicated.

Now, if we take the *symbol level* to be more or less as described above, and if we identify the principles that govern this level with some very general properties of what I have been calling “functional architecture,” we will find that the existing (empirically observed) functional architecture of humans, and perhaps of other creatures, is compatible with a wide range of processes and regularities that do not occur in human cognition. Some of these processes and regularities may not occur simply because the right conditions do not happen to be present, that is, they *could* occur within some class of variation of such conditions—for example, if the cultural or social context were substantially different (here, the relevant class of variations is an important consideration, one we will take up in detail in chapter 5). On the other hand, there are other types of processes that would not occur under this class of variation of antecedent conditions, though conceivably they could be caused by brain damage or similar noncognitive intervention. What are some of these processes? I suggest below that they are processes that are characterizable as *semantically deviant*.

Even with no detailed theory of human functional architecture, we already see that principles of the architecture can only constrain transitions among representational states in certain ways, specifically in certain structural ways that are *independent of the content* of these representational states. For example, limitations in the size of working memory may constrain the number of items that can be interrelated at one time, or they may necessitate certain recodings that lead to certain memory-time tradeoffs (see, for instance, the proposal in Newell,

1973b). It is clear that whatever constraints there may be in detail, the functional architecture permits application of arbitrary rules governing transitions among functional states so long as the rules conform to certain very general structural or functional principles. There is reason, though, to think there are other sorts of principled constraints not specifiable in terms of properties of information-processing mechanisms that constrain rules in certain cases, including some of the clearest cases of what might be called "high-level cognitive processes."

As far as the constraints of the functional architecture are concerned, rules are permitted that would be viewed as nonsensical or bizarre or incoherent by someone who interpreted the functional states as having representational content. For example, there could be a regularity among functional states that would be interpreted as the rule that if one believes all ravens are black, and that there is a raven in the tree, then that person goes into a state corresponding to the belief that the sky is falling; or there might be a regularity that, under the representational interpretation, would come out as the rule that if someone tells you a building is on fire, you come to believe that $2 + 3 = 12$.

Clearly, nobody would choose to interpret the functional states that way, and no empirically adequate model would contain such rules, since the rules would not generate the right behaviors. Whether anyone would ever hypothesize such rules is not the point. The question still arises in a cognitive system—What principle prevents them from occurring?—as it does in the case of the biological examples discussed. Why don't certain rules permitted by the constraints of the functional architecture, in fact, occur? More to the point, since the absolute non-occurrence of such cases is not what is crucial, what distinguishes one kind of rule (the incoherent or nonsensical ones) from the other kind (the coherent or sensible ones), when both kinds are permitted within the constraints of the functional architecture? As in the example from biology, there are two possible answers. One is that the nonoccurrence of one kind of rule may be due to the initial state of the world from which organisms evolved; hence, the distinction between the two kinds of rules could be considered an accident, since the distinction between the kinds of rules that occur and those that do not occur could, in principle, have been quite different. The other answer is that there may be additional principles, perhaps universal in human cognitive systems, not stateable in terms of constraints on the functional mechanisms. (I will have more to say about the importance of distinguishing different sources of constraints in chapter 7, since that is one of the main issues raised by the proposal that certain kinds of cognitive processes are *analogue*.)

To say that the distinction between the two kinds of rules is due to

the survival advantage of having truth-preserving transformations (that is, valid inferences) may be correct—though it is easy to overestimate the explanatory value of such appeals to evolution—but it does not diminish the present point: that principles must be imported from outside the symbol level, or the level of functional architecture, to explain these particular constraints, and that these principles must be stated in terms of what the functional states represent, here, in terms of “truth preservation.”⁷ Although many doubt the possibility of couching the principle of rationality in terms of survival value, the present point is neutral on this question.

We thus come once again to the view that, in a cognitive theory, the reason we need to postulate representational content for functional states is to explain the existence of certain distinctions, constraints, and regularities in the behavior of at least human cognitive systems, which, in turn, appear to be expressible only in terms of the semantic content of the functional states of these systems. Chief among the constraints is some principle of rationality. It is also a consequence of this view that deviations from rationality can arise because of interactions of intentional-level (or knowledge-level) regularities with functional-level constraints (that is, constraints of the functional architecture), just as certain functional regularities may not be realized because of intrusions of physical-level properties (such as when computer program failures are due to faulty electronic components). Effects can penetrate upward through levels, since each level is supervenient on levels below; that is, there can be no differences at level n unless there is some difference at level $n-1$, even though the converse is not true (because of the multiple-instantiation property of ascending levels; supervenience of psychological states on biological states entails that there cannot be two different thoughts unless there are *some* biological differences between the two underlying brain events).

Where Does the Semantic Interpretation Come From?

It appears that we need to appeal to the content of representations in

7. It is one of the tasks of modern logic to show that to some extent—at least, within deductive logic—it is possible to specify such principles as validity (surely a constitutive property of rational thought) purely in formal or syntactic terms, that is, without reference to the content of nonlogical terms. Even if all rational rules could be specified in this way, the point at issue here would be unaffected. Every rational rule might be specified in a formal notation, and the set of such rules that characterize an ideally rational individual might be listed; but one could still not express syntactically what it was that all such rules have in common which distinguishes them from other syntactically well-formed rules—because *that* is a semantic property (a property that cannot be stated without reference to what is represented by the expressions in the rules).

order both to account for certain kinds of generalizations and to give a principled account of certain constraints and capacities, such as those embodied in the principle of rationality. It also appears that, in a certain sense, it may be possible to accommodate this need within a functional framework by providing an interpretation of inputs and outputs and of at least some states of the model. Doing so, however, does not solve the entire problem, for it might still be argued that such interpretations are not the model's interpretations, that they are merely attributions made by the theorist. It does matter to us whether the states are representational for the model or merely for us. This is a question of whether the semantics of the states are original—as the semantics of our thoughts presumably are—or whether they are derived, as is the case with books. The meanings books have are in a derived sense; they mean something *only for us* because their authors intend them to have certain meanings when they write them. In cognitive science, however, we want something stronger than derived semantics, inasmuch as we want to explain the system's behavior by appealing to the content of its representations.

A dilemma is presented here. As we have already noted, the semantics of representations cannot literally cause a system to behave the way it does; only the material form of the representation is causally efficacious. When we say that we do something because we desire goal *G* or because we believe *P*, we are attributing causality to a representation, though it is not the content that does the causing. As we have noted, what is being represented need not exist, and even when it does, its only relevant aspect is the way it is conceptualized. What actually do the causing are certain physical properties of the representational state—but in a way that reflects the representational state's content. There is, if you like, a parallel between the behavioral patterns caused by the physical instantiation of the representational states and the patterns captured by referring to the semantic content of these states. How can this be? How can such a parallel be maintained coherently over time?

Only one nonquestion-begging answer to this dilemma has ever been proposed: that what the brain is doing is exactly what computers do when they compute numerical functions; namely, their behavior is caused by the physically instantiated properties of classes of substates that correspond to *symbolic codes*. These codes reflect all the semantic distinctions necessary to make the behavior correspond to the regularities that are stateable in semantic terms. In other words, the codes or symbols are equivalence classes of physical properties which, on one hand, cause the behavior to unfold as it does, and on the other, are the bearers of semantic interpretations that provide the needed

higher-level principle for their individuation and for stating the generalizations. As dyed-in-the-wool realists, we propose as the next step exactly what solid-state physicists do when they find that postulating certain unobservables provides a coherent account of a set of phenomena: we conclude that the codes are “psychologically real,” that the brain is the kind of system that processes such codes and that the codes do in fact have a semantic content.

Many people are quite familiar with this idea, though it remains fairly controversial outside cognitive science. The uneasiness this causes in some people is absent among those who work with computer models, because this story is precisely, and literally, what happens in computers—with one difference. The states of a computer can be given many different semantic interpretations; indeed, the same symbolic states are sometimes interpreted as words, sometimes as numbers, chess positions, or weather conditions. In the case of a psychological model we want to claim that the symbols represent one content and no other, since their particular content is part of the explanation of its regularities. Thus, even if we accept the “syntax parallels semantics” view, there remains the question, What determines what a given (syntactically articulated) state represents? I shall examine several proposals for dealing with this problem that have surfaced over the years and then offer some extremely modest proposals for thinking about this problem in its relation to computational theories. My proposals are incomplete as answers to the puzzle because the entire problem of semantics is, in general, poorly understood. The proposals are intended only to suggest that the lack of a definitive answer need not be taken as an indictment of the cognitive-science enterprise, since no proposal concerning mental content is free of all problems.

Early versions of mentalism, all of which place conscious experience in a central methodological position, had an answer to the question, What causes certain mental events to have certain contents? In these versions, at least some mental contents represent certain things because they *resemble* them. An image of *X* represents *X* precisely because the conscious mental representations, or images, *look like X*. Such a view probably is not far from the common notion of visual imagery. If you were to ask a group of people how they know their image of a duck actually represents a duck, rather than, say, a rabbit, they might reply that the image looks like a duck. For several reasons, however, this answer does not explain why the image is a representation of a duck. For example, even in the introspectionist approach, the image need not closely resemble a duck for people to take it as a duck since it is *their* image, they can take it as virtually anything they wish; after all, the word *duck* refers to a duck without in any way resembling a

duck. As Wittgenstein points out, the image of a man walking up a hill may look exactly like the image of a man walking backward down a hill; yet, if they were my images, there would no question of their being indeterminate—I would know what they represented.

The relation of resemblances is not well defined. Whether one thing resembles another is not a physically (or geometrically) definable property; resemblance depends on what the viewer knows or believes. To me, most birds closely resemble one another, but to a birdwatcher friend they are as different as ducks and rabbits. Resemblance cannot be specified except in relation to a viewer, which, in the case of mental contents, amounts to positing an intelligent, knowledgeable, problem-solving homunculus. Resemblance provides no basis for specifying the semantic content of mental representations. In any case, this kind of mentalism has little to say about the second part of the issue of representation: how behavior can be caused by the property that gives them their content—which, in this case, is resemblance. One can have something called “association of ideas,” and one can apply consciously memorized rules; but there is no room in this mentalism for appeal to consciously unavailable, content-dependent processes. Even with memorized rules there is the problem of who (what sort of mechanism or homunculus) applies them. This problem, in fact, formed part of the criticism frequently leveled at early mentalistic cognitivism: that these theories “left the organism lost in thought,” with no way to generate behavior from their stock of images without the intervention of a homunculus.

In addition, certain schools of behaviorism have a proposal for the first part of the task of explicating representational content. Mediational behaviorists (especially Osgood, 1963) and certain speculative neurophysiologists (for example, Hebb, 1949) take the position that a brain event can be said to represent something if that event is sufficiently like (possesses a subset of the properties of) the event that takes place when that something actually is perceived. Another, more radically behaviorist version requires that the mediational event evoke an internal “preparatory response” (as in Hull’s “anticipatory goal response”) that is sufficiently like the response that would have been evoked by the corresponding stimulus. Neither position is satisfactory, because of the properties of representations we have already noted (for example, we can think about objects we have neither perceived nor have any disposition to behave toward, such as, perhaps, quarks). In any case, the only mechanism behaviorism provides for explicating the representing relation is that of association. Association, in turn, must be established by such principles as contiguity and evoked by the activation of other associated items (otherwise we would not have provided the naturalistic

account of the semantics of the functional states sought by behaviorism). A chain of contiguous events mediating between a brain state and an object, however, cannot form the basis of representation, for reasons discussed above, namely, that it is neither necessary nor sufficient that the state of an organism be linked by a series of contiguous events to the object that the state represents. Not only can I think of things to which I obviously am not in this sort of relation (for example, nonexistent things), but when I do think of X, I do not thereby think of associates of X; indeed, I need not think of properties that are necessarily coextensive with X, such as shape, size, weight, color, and so on.

The basic problem is that *representing* is a semantic relation, that semantic relations, like logical relations, appear not to be causally *definable*, at least not in any direct way (see Fodor, 1980a) even though they may on various token occasions be established by or realized in various, different, causally explainable ways. It may be that representation requires that certain causal relations with the world were established at some point in the organism's history. For example, in order to represent the sensory feature *red* as a particular sensory quality, some aspect of the representational state must be causally related to an occasion on which the organism was in sensory contact with a color in the category red. However, the general attempt to define or reconstruct semantic relations in terms of associatively established links—or any other straightforward series of causal events—is bound to encounter difficulty. It is like all the other cases of multiple-instantiation relations we encountered in our discussions—though each case of a semantic relation may be accompanied by *some* causal relation, the two are not type equivalent.

In contrast with naturalistic approaches to representation, at least one version of the functionalist approach (the “coherence view,” popular with computationalists) does not attempt to specify the semantics of functional states in terms of a purely causal link between the functional state and the object represented. Rather, the relation of representing is taken as arising from the role of the functional state within the complete theory, including the way the organism interacts with its environment through mechanisms called transducers, the way it relates to those of its inputs and outputs that are considered to be interpreted (which, for humans, is clearest in the case of the linguistic signals they receive and emit) and the relation of the organism's behavior and its own history of such interactions.

Although, in the computational model, the symbolic codes themselves do not specify their intended interpretation (and the model's behavior is not influenced by such an interpretation), the cognitive theory that makes claims about what it is a model of, which aspects of it are

supposed to model something and which are not, *does* have to state what the states represent, for reasons already noted, having to do with explanation and with capturing generalizations. The cognitive theory would be gratuitous, or, at best, weakly equivalent or “mere mimicry,” if the ascription of some particular representational content to states of the model were not *warranted*. The particular interpretation placed on the states, however, appears to be extrinsic to the model, inasmuch as the model would behave in exactly the same way if some other interpretation had been placed on them. This observation is what has led some (for example, Searle, 1980) to claim that functional models (at least of the computational type) do not actually contain representations, that it is we, the *theorists*, who, for reasons of convenience, hold that functional states in computer models have representational content—and that adds up to what Searle (1980) calls *weak artificial intelligence*, that is, a functionalist account in which formal analogues “stand in” for but neither have nor explain the representational (or intentional) contents of mental states.

Notice that if this view were correct, it would undermine my claim that functional models, at least of the computational variety, address the issue of representation-governed behavior. That is because I want to maintain (see chapter 4) that computation is a literal model of mental activity, not a *simulation* of behavior, as was sometimes claimed in the early years of cognitive modeling. Unlike the case of simulating, say, a chemical process or a traffic flow, I do not claim merely that the model generates a sequence of predictions of behavior, but rather that it does so in essentially the same way or by virtue of the same functional mechanisms (not, of course, the same biological mechanisms) and *in virtue of* having something that corresponds to the same thoughts or cognitive states as those which govern the behavior of the organism being modeled. Being the same thought entails having the same semantic content (that is, identical thoughts have identical semantic contents).

Now, Searle is surely right in saying that it does not follow merely from the fact that computational models produce the same behavior, and that their states are given a certain semantic interpretation, that such systems work in the same way as humans do. By the same token, it does not follow from the fact that it was the theorist who provided the interpretation of the symbols that such an interpretation is simply a matter of convenience or that there is a sense in which the interpretation is the theorist’s rather than the system’s. Much depends on the theorist’s reasons for giving that particular interpretation. In my view, the question of whether the semantic interpretation resides in the head of the theorist or in the model itself is the wrong question to ask. A better question is: What fixes the semantic interpretation of

functional states? or, What latitude does the theorist have in assigning a semantic interpretation to the states of the system?

Admittedly, in many models that take the form of computer programs, there is much latitude in assigning semantic interpretations to the symbols, hence, to the model's states. Indeed, we routinely change our interpretation of the functional states of a typical computer, sometimes viewing the states as numbers, sometimes as alphabetic characters, sometimes as words or descriptions of a scene. Even when it is difficult to think of a coherent interpretation different from the one the programmer had in mind, such alternatives are, in principle, always possible. (There is an exotic result in model theory, the Lowenheim-Skolem theorem, which guarantees that such programs can always be coherently interpreted as referring to integers and to arithmetic relations over them.) If, however, we equip the programmed computer with transducers so it can interact freely with a natural environment and a linguistic one, as well as the power to make inferences, it is far from obvious what if any latitude the theorist (who knows how the transducers operate and therefore what they respond to) would still have in assigning a coherent interpretation to the functional states so as to capture psychologically relevant regularities in behavior. If the answer is that the theorist is left with no latitude beyond the usual inductive indeterminism all theories have in the face of finite data, it would be perverse to deny that these states had the semantic content assigned to them by the theory.

Suppose such constraints (and, possibly, others not yet considered) fixed the possible interpretations that could be placed on representational states. Would that resolve the argument concerning whether we are warranted in ascribing a particular semantic content to these states? Here, I suspect that one would run into differences of opinion that could prove unresolvable. There arises immediately the question whether such functional models are fundamentally different from us, inasmuch as we alone possess special, privileged access to the content of our thoughts, presumably something our model would lack unless it, too, was conscious. Even if we have strong intuitions concerning such questions, many of us believe that such intuitions should be considered no more than secondary sources of empirical constraint, whose validity should be judged on the basis of how successful the resulting theoretical systems are in accounting for all aspects of the phenomena in question. Information-processing theories have achieved some success in accounting for such aspects as problem-solving, language-processing, and perception, by deliberately glossing over the conscious-unconscious distinction—by grouping under common principles processes necessary to account for functional capacities and for patterns

of behavior, independent of whether people are aware of the underlying processes and representations. At the same time the theories have set aside questions about what constitutes qualia, or “raw feels”—dealing only with some of their reliable functional and semantic correlates (for example, the *belief* that one is in pain, as opposed to the *experience* of the pain)—and have to a large extent deliberately avoided the question of what gives functional states their semantics, opting instead to assign such semantic interpretations according to a principle of global coherence and waiting to see whether the resulting theory lives up to the usual canons of scientific performance.

This is about as much as I need say about semantic level and representations. After completing the tour of the cognitive-science program in the next six chapters, we will return to reconsider the fundamental proposal that human cognition and certain computer systems share the remarkable feature that they are what Dennett (1978) calls “semantic engines” or what George Miller (1984) has referred to as “informavores.” At this point I have introduced computers, as it were, through the back door; I have been using the idea of symbol system without properly introducing computers and computation. Because computing plays a major role in the cognitive science project, in the next chapter I take a step backward and examine the symbol-system phenomenon in some detail.

End Note 1: When Are Functional States Representations?

It is not straightforward to give either necessary or sufficient conditions for when a system can be said actually to be governed by internal representations. These conditions, it should be appreciated, are rarely available even for the clearest distinctions; no unproblematic defining conditions are available for such basic notions as cause, explanation, and cognition or for most other fundamental concepts in science. Yet, in all such cases there are usually good indications (at least, for most clear cases) of when the concepts are applicable. So, too, are there indicators of when the explanation of certain phenomena cannot avoid appeal to representations without losing important principles and generalizations. As is shown by many of the examples we have discussed, there are clear cases where the relation between stimulus and response (again, under the appropriate cognitive description) is itself unsystematic in general, yet where the relation between a certain interpretation (hence, representation) of the stimulus and a certain interpretation of the response is not only systematic but may even be seen as rational. In general, we find that when we have a situation in which appealing to the semantic content of functional states enables us to explain regularities

that would otherwise go unexplained, it is because the relation between the ascribed content and the functional states is not a simple one-to-one correspondence. Typically, we get only an explanatory advantage by appealing to representational content, when the semantic interpretation *S* we give to certain states *I* happens to have such characteristics as the following:

(a) According to our method of assigning content there will more than one distinct *I* in our model that corresponds to some actual or possible *S*; that is, there will, in general, be more than one distinguishable way of representing something in our scheme).

(b) According to our method of assigning content, *I*'s will also occur in our model for which no corresponding *S* exists—or is even possible; that is, according to our theory, the system can, in general, represent states of affairs that are contrary to fact.

(c) Given our content-assigning scheme and the functional model that accompanies it, there may also be an unlimited number of ways for the system to come to be in state *I* other than as a direct causal consequence of the occurrence of the corresponding state of the environment identified as *S*. Here, a representational account might be able to show how these different ways of achieving state *I* are related under the appropriate semantic interpretation. In particular, we may be able to explain the transition from *I'* to *I* by showing that under the particular interpretation we have selected, the transition instantiates some logical, heuristic, or associative rule.

(d) Closely related to point (c), we may also find that the state-to-state transition regularities that do occur in certain circumstances can be distinguished from other possible transitions that do not occur, by viewing the individual states *I* as representing some *S* and by appealing to regularities that exist in the domain of *S*, including purely conceptual ones. We encounter excellent examples of this case in discussing models of mental imagery in chapter 8.

Again, it must be emphasized that these are neither necessary nor sufficient conditions for a system to be genuinely representational. The criterion for their being genuinely representational remains whether generalizations are missed when one is denied a representational vocabulary that refers to objects and properties represented by the system and whether there are principled constraints on the system's behavior that can be captured only using such semantic principles as rationality. Nonetheless, various conditions such as those sketched above are among the conditions that provide *prima facie* indications that certain functional states should be viewed as semantically interpreted, since such conditions are symptomatic of an independence of the state's role from its causal antecedents and of a dependence of its role on its semantic

content, that is, on what it represents. For example, these conditions entail that the way cognitive, or representational, processes unfold has a high degree of independence from the organism's causal interactions with the world. They separate the content of the representation from the stimulus conditions by an act of interpretation or encoding, which itself may involve an act of inference, as opposed to instantiating a natural, causal law. As Dretske (1981, p. 188) puts it, an important property of a representation is "its muteness about the particular manner in which the information (constituting its semantic content) arrived." What regularities follow from the presence of a certain stimulus depend not only on what the stimulus was but on what it was taken to be (on what it was "seen as"). The latter, in turn, depend on the system's other semantically interpreted structures (beliefs, goals, fears, imagination).

Conditions (c) and (d) arise because a representation that serves an ineliminable role in the explanation of a system's behavior is generally one that enters into logical (or, at least, quasi-logical) relations, as well as other relations governed by rules and arbitrary conventions, stated in terms of the domain represented. This helps to explain how organisms appear capable of anticipating novel situations, and of explaining what appears to be suboptimal behavior—behavior that occurs when the organism has false beliefs. Because precisely the same event or object can be represented in different ways, the system, or organism, can behave differently with respect to the object on different occasions or in different "states of knowledge." All are characteristics of intentional (or epistemic) or representational states (states individuated by their content or by their semantic properties, which causes them to differ substantially from functional states in general, that is, states individuated in terms of their role in generating different behaviors).

Note that the characteristics of representations discussed above are not present in any examples of purely functional explanations mentioned earlier, including cases in which the explanation adverts to a design vocabulary. Clearly, these are not characteristics of the functional states of television sets or watches. Nor are they true of neural states, that is, states individuated in terms of their neurophysiological properties. Although certain neural states could be described as representing the state of the retina, the motor system, or some other part of the nervous system (as P. S. Churchland, 1980, does), the relation between the neural state in question and what it is taken to represent fails to meet numerous conditions on representation suggested above; for example, transitions among them cannot be described in terms of such content-dependent rules as inference. These relations are purely causal and must be sharply distinguished from *semantic* relations. Describing a

certain neural pattern as “representing” the position of a limb is exactly the same as describing the pattern of gears in a clock as representing the position of the hands on the face of the clock or of the time of day: both are a derivative senses of *representing* which gains us no explanatory advantage over a neurophysiological or mechanical description.

End Note 2: On the Notion of Representational Content

I have been speaking of representational content as though it were a simple, straightforward notion. Although the points made in this chapter do not depend crucially on precisely how content is understood, the idea of content does raise some thorny philosophical problems. First, it should be kept in mind that content refers to the *conceptual* content of representations (thoughts, images, goals) and not to the reference of these representations. The contents of representations are not actual objects in the world. For one thing, most of our mental contents have no referents in the world. This is patently true of goals, since goals describe nonactual states of affairs, and it is also true of many of the fears and beliefs that motivate us. Moreover, even when mental representations succeed in referring, what they in fact refer to is not what is relevant. The grain of mental contents is not the same as the grain of distinct, objective states of the world; precisely the same state of the world can correspond to numerous, different conceptual contents. A thought about the person who just drove up in a car has a different content from a thought about my daughter, even though, as it happens, the person who just drove up in my car is my daughter. A desire to marry the girl who lives on Piccadilly Street has a quite different content from the desire to marry Mary Jones, whether or not Mary Jones is the girl on Piccadilly Street toward whom both desires are directed. Oedipus had no desire to marry his mother, even though the person he wished to marry was, in fact, his mother, as he later tragically discovered. In all cases the two contents have different meanings, or what Frege calls different senses, though they refer to the same actual states of affairs or to the same individuals. They are said to be intensionally distinct though extensionally equivalent. Thus we see that we want to count two token mental states (thoughts and goals) as having the same content only if they are intentionally the same.